1ª parte

Escolha múltipla: C - D - C - A - A

6. $y(t) = -\frac{1}{4t^2} + ct^2$, $c \in \mathbb{R}$ e $D = \mathbb{R} \setminus \{0\}$

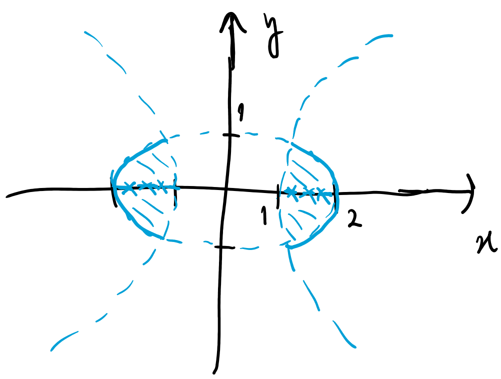
7. a) Modelo $\begin{cases} y' = k(T-28) \\ y(0) = 3 \\ y(50) = 8 \end{cases}$

Solução $y(t) = 28 - 25\, e^{\frac{\log(4/5)}{50} t}$.

b) $t = 50 \dfrac{\log(3/25)}{\log(4/5)}$.

8

a) $D = \{(x,y) \in \mathbb{R}^2 : \frac{x^2}{4} + y^2 \leq 1 \wedge x^2 - y^2 \geq 1 \wedge x \neq 0 \wedge y \neq 0\}$



b)

$\text{int}(D) = \{(x,y) \in \mathbb{R}^2 : \frac{x^2}{4} + y^2 < 1 \wedge x^2 - y^2 > 1 \wedge x \neq 0 \wedge y \neq 0\}$

$\text{fr}(D) = \{(x,y) \in \mathbb{R}^2 : (x = 0 \wedge \frac{x^2}{4} + y^2 \leq 1 \wedge x^2 - y^2 \geq 1) \vee$

$$f_2(D)=\{(x,y)\in\mathbb{R}^2 : \left(x=0 \wedge \frac{x^2}{4}+y^2 \leq 1 \wedge x^2-y^2\geq 1\right) \vee$$

$$\left(y=0 \wedge \frac{x^2}{4}+y^2\geq 1 \wedge x^2-y^2\geq 1\right) \vee$$

$$\left(x^2-y^2=1 \wedge \frac{x^2}{4}+y^2 \leq 1\right) \vee$$

$$\left(\frac{x^2}{4}+y^2=1 \wedge x^2-y^2\geq 1\right)\}$$

9 b) $\frac{\partial g}{\partial x}(0,0)=0$ e $\frac{\partial g}{\partial y}(0,0)=0$

$g'_u(0,0)=\frac{1}{2}$

e) Não porque $g'_{\vec{u}}(0,0)\neq \nabla g(0,0)^T\cdot u$

2ª parte

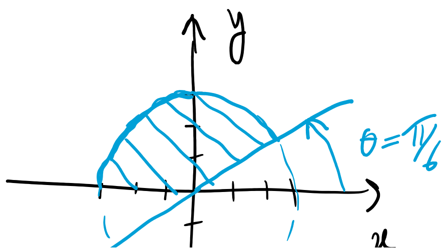
Escolha múltipla: A - B - B - B - D

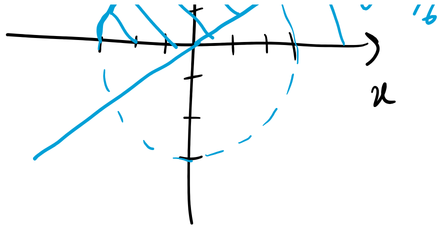
6. $\left(\frac{1}{9}, \frac{1}{6}, \frac{1}{3}\right)$

7. b) $\nabla\phi(0,\pi/2)=\begin{bmatrix}0\\0\end{bmatrix}$

c) $\phi'_{(3,-2)}(0,\pi/2)=0$

8 a)





b) $\frac{15\pi}{4}$

9 a) $S=\{(r,\theta,z)\in \mathbb{R}^+\times[0,2\pi[\times\mathbb{R}:\ 0\le r\le 1 \wedge \sqrt{3}r\le z\le\sqrt{4-r^2}\}$

b) $\frac{8}{3}(2-\sqrt{3})$.